

1 Drivetrain Sizing and Motor Selection

1.1 Design Objective

The drivetrain is responsible for providing the required traction, speed, and torque for the differential-drive AMR-X platform. Since the initially selected Maxon EC 45 Flat 70 W motor and GP 42C 33:1 gearbox were selected before a complete drivetrain calculation was performed, a preliminary sizing study was conducted to verify their compatibility with the robot requirements.

The objective of this study is to determine the approximate continuous torque, speed, and power required at each drive wheel, and subsequently identify suitable brushless DC (BLDC) motor and gearbox combinations.

The calculations below are based on the current AMR-X mechanical requirements and several preliminary engineering assumptions. These assumptions should be validated and refined during the detailed mechanical design stage.

1.2 Preliminary Design Assumptions

The following parameters are used for the initial drivetrain sizing:

Parameter	Value
Maximum unloaded robot mass	80 kg
Rated payload	50 kg
Design mass	130 kg
Drive architecture	Differential drive
Number of driven wheels	2
Initial wheel diameter	180 mm
Wheel radius	90 mm
Normal operating speed	0.8–1.2 m/s
Maximum speed	1.5 m/s
Maximum slope	5°
Preliminary acceleration	0.3 m/s ²
Rolling resistance coefficient	$C_{rr} = 0.02$
Preliminary safety factor	1.5

The design mass is conservatively taken as the maximum unloaded mass plus the rated payload:

$$m = 80 + 50 = 130 \text{ kg} \quad (1)$$

The acceleration and rolling resistance values are preliminary assumptions used to obtain an initial engineering sizing point. They are not yet formal system requirements.

1.3 Wheel Speed Calculation

The selected initial drive wheel has a diameter of:

$$D = 0.18 \text{ m} \quad (2)$$

The wheel circumference is therefore:

$$C = \pi D \quad (3)$$

$$C = \pi(0.18) \approx 0.565 \text{ m} \quad (4)$$

The required wheel rotational speed for a given vehicle velocity is:

$$N_w = \frac{60v}{\pi D} \quad (5)$$

The resulting wheel speeds are:

Robot velocity	Wheel speed
0.8 m/s	84.9 rpm
1.0 m/s	106.1 rpm
1.2 m/s	127.3 rpm
1.5 m/s	159.2 rpm

Therefore, the drivetrain must provide approximately 85–159 rpm at the drive wheels.

1.4 Traction Force Requirement

The total traction force is estimated from three main contributions:

$$F_{total} = F_{acc} + F_{slope} + F_{rolling} \quad (6)$$

where:

- F_{acc} is the force required for vehicle acceleration,
- F_{slope} is the force required to overcome gravity on the slope,
- $F_{rolling}$ is the rolling resistance force.

1.4.1 Acceleration Force

Using the preliminary acceleration assumption:

$$a = 0.3 \text{ m/s}^2 \quad (7)$$

the acceleration force is:

$$F_{acc} = ma \quad (8)$$

$$F_{acc} = 130(0.3) = 39 \text{ N} \quad (9)$$

Therefore:

$$\boxed{F_{acc} = 39 \text{ N}} \quad (10)$$

1.4.2 Slope Force

For a maximum slope of:

$$\theta = 5^\circ \quad (11)$$

the gravitational component acting along the slope is:

$$F_{slope} = mg \sin(\theta) \quad (12)$$

Using $g = 9.81 \text{ m/s}^2$:

$$F_{slope} = 130(9.81) \sin(5^\circ) \quad (13)$$

$$\boxed{F_{slope} \approx 111.2 \text{ N}} \quad (14)$$

1.4.3 Rolling Resistance

The rolling resistance is approximated by:

$$F_{rolling} = C_{rr}mg \cos(\theta) \quad (15)$$

with:

$$C_{rr} = 0.02 \quad (16)$$

Therefore:

$$F_{rolling} = 0.02(130)(9.81) \cos(5^\circ) \quad (17)$$

$$\boxed{F_{rolling} \approx 25.3 \text{ N}} \quad (18)$$

1.4.4 Total Traction Force

The total estimated force is:

$$F_{total} = 39 + 111.2 + 25.3 \quad (19)$$

$$\boxed{F_{total} \approx 175.5 \text{ N}} \quad (20)$$

1.5 Wheel Torque Requirement

The required torque at the ground for the two drive wheels is:

$$T_{total} = F_{total}r \quad (21)$$

where:

$$r = 0.09 \text{ m} \quad (22)$$

Thus:

$$T_{total} = 175.5(0.09) \quad (23)$$

$$T_{total} \approx 15.8 \text{ N m} \quad (24)$$

Assuming equal load sharing between the two driven wheels:

$$T_{wheel} = \frac{15.8}{2} \quad (25)$$

$$\boxed{T_{wheel} \approx 7.9 \text{ N m}} \quad (26)$$

Applying the preliminary safety factor:

$$T_{design} = 1.5T_{wheel} \quad (27)$$

$$T_{design} = 1.5(7.9) \quad (28)$$

$$\boxed{T_{design} \approx 11.9 \text{ N m/wheel}} \quad (29)$$

Therefore, the preliminary drivetrain target is approximately 11.9 N m of continuous design torque per drive wheel.

1.6 Mechanical Power Requirement

The mechanical power required at the wheels can be estimated using:

$$P = Fv \quad (30)$$

At the maximum vehicle velocity of 1.5 m/s:

$$P_{wheel,total} = 175.5(1.5) \quad (31)$$

$$\boxed{P_{wheel,total} \approx 263 \text{ W}} \quad (32)$$

For two driven wheels:

$$P_{wheel} \approx \frac{263}{2} \quad (33)$$

$$\boxed{P_{wheel} \approx 131.5 \text{ W}} \quad (34)$$

Assuming an approximate gearbox efficiency of 85%, the corresponding motor-side mechanical power is approximately:

$$P_{motor} = \frac{131.5}{0.85} \quad (35)$$

$$\boxed{P_{motor} \approx 155 \text{ W/wheel}} \quad (36)$$

Thus, a preliminary target of approximately 155 W continuous mechanical power per motor is obtained for the assumed worst-case operating point.

This value is a sizing reference rather than a final motor power requirement, since the actual acceleration and operating speed on the maximum slope have not yet been formally specified.

2 Evaluation of the Initially Selected EC 45 + GP 42C

2.1 Initial Configuration

The initial drivetrain selection consisted of:

- Maxon EC 45 Flat BLDC motor, 70 W, 24 V;
- Maxon GP 42C planetary gearbox;
- Gearbox reduction ratio: 33 : 1;
- Hall sensor feedback.

The EC 45 70 W motor provides approximately 128 mN·m continuous torque at 24 V. Its nominal speed is approximately 4,750–4,860 rpm, depending on the specific datasheet configuration, with a no-load speed around 6,110 rpm.

2.2 Torque Evaluation

Using the motor's continuous torque:

$$T_m = 0.128 \text{ N m} \quad (37)$$

and an assumed gearbox efficiency of approximately:

$$\eta_g = 0.85 \quad (38)$$

the estimated continuous gearbox output torque is:

$$T_{out} = T_m G \eta_g \quad (39)$$

$$T_{out} = 0.128(33)(0.85) \quad (40)$$

$$\boxed{T_{out} \approx 3.59 \text{ N m}} \quad (41)$$

This is substantially lower than the preliminary design target:

$$T_{design} \approx 11.9 \text{ N m} \quad (42)$$

Therefore:

$$\boxed{\text{EC 45 + GP 42C 33:1 does not satisfy the preliminary torque requirement.}} \quad (43)$$

2.3 Power Evaluation

The EC 45 has a nominal mechanical power of approximately:

$$P_m = 70 \text{ W} \quad (44)$$

whereas the preliminary calculation indicates approximately:

$$P_{motor} \approx 155 \text{ W/wheel} \quad (45)$$

for the assumed maximum-load, maximum-slope, acceleration condition.
Therefore:

The 70 W EC 45 is also insufficient in continuous power for this design point.

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2.4 Can a Higher Gear Ratio Solve the Problem?

A higher gearbox ratio increases the available wheel torque, but it simultaneously reduces wheel speed for a given motor speed. It also cannot increase the mechanical power available from the motor.

The approximate gearbox ratio required to obtain the design torque from the EC 45 is:

$$G_{required} = \frac{T_{design}}{T_m \eta_g} \quad (47)$$

$$G_{required} = \frac{11.9}{0.128(0.85)} \approx 109 \quad (48)$$

Therefore, a ratio of approximately 109 : 1 would theoretically be required based on continuous torque.

However, at the maximum wheel speed of approximately 159 rpm, this would require:

$$N_m = 159(109) \quad (49)$$

$$N_m \approx 17,300 \text{ rpm} \quad (50)$$

which is far beyond the EC 45's operating speed range.

Consequently, increasing the gearbox ratio cannot solve both the torque and speed requirements.

The EC 45 motor itself must be replaced for the current design assumptions.

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3 Alternative Motor and Gearbox Candidates

The replacement candidates are selected according to the following criteria:

- 24 V operation;
- BLDC motor technology;

- Hall sensor availability for electronic commutation;
- Sufficient continuous torque and power;
- Suitable motor speed for the required wheel speed;
- Compatibility with the EPOS4 motion controller;
- Availability of a suitable planetary gearbox;
- Industrial-quality construction suitable for a manufactured V1 prototype.

An important architectural point is that the motor itself does not communicate through CANopen. CANopen communication is handled by the EPOS4 motor controller. The EPOS4 operates as a CiA 402 CANopen motion-control slave and can control BLDC motors using Hall sensors and encoder feedback. Therefore, all candidates below can retain the planned:

$$\boxed{\text{STM32} \rightarrow \text{CANopen} \rightarrow \text{EPOS4} \rightarrow \text{BLDC motor}} \quad (52)$$

architecture, provided the exact EPOS4 variant is correctly selected for the motor's electrical requirements.

3.1 Candidate 1: Maxon EC 60 Flat 200 W

The Maxon EC 60 Flat 200 W is a 24 V brushless motor equipped with Hall sensors. According to the manufacturer, the motor has a nominal speed of approximately 3240 rpm, a no-load speed of approximately 4300 rpm, and a maximum continuous torque of approximately 540 mN·m. The motor is rated at 200 W and has a maximum efficiency of approximately 85%.

Parameter	Specification
Motor	Maxon EC 60 Flat
Nominal voltage	24 V
Continuous power	200 W
Continuous torque	0.540 N·m
Nominal speed	3240 rpm
No-load speed	4300 rpm
Continuous current	9.34 A
Feedback	Hall sensors
Motor diameter	60 mm
Motor mass	Approximately 360 g

A possible gearbox ratio must be selected carefully. For example, with a 26 : 1 reduction and approximately 83% gearbox efficiency:

$$T_{out} = 0.540(26)(0.83) \quad (53)$$

$$\boxed{T_{out} \approx 11.65 \text{ N m}} \quad (54)$$

This is close to the preliminary 11.9 N·m design target.

However, the corresponding motor speed at the maximum vehicle velocity would be:

$$N_m = 159.2(26) \quad (55)$$

$$\boxed{N_m \approx 4140 \text{ rpm}} \quad (56)$$

which is below the motor's approximately 4300 rpm no-load speed.

The EC 60 is therefore a promising candidate, although the detailed motor speed–torque operating point and gearbox continuous torque rating must be verified before final selection.

Advantages:

- 200 W continuous power provides a substantially larger power margin than the original EC 45;
- 0.540 N·m continuous motor torque;
- Compact 60 mm diameter;
- 24 V operation compatible with the proposed AMR electrical architecture;
- Hall sensors available for BLDC commutation;
- Retains compatibility with the planned EPOS4/CANopen architecture.

Disadvantages:

- The motor is already close to its maximum speed at the required 1.5 m/s wheel speed when using a high reduction ratio;
- The exact gearbox/motor combination must be checked for continuous output torque;
- Active ventilation is used on the selected 200 W version and must be considered during mechanical integration.

3.2 Candidate 2: Maxon EC 90 Flat 220 W

The Maxon EC 90 Flat 220 W provides a higher continuous torque and power while remaining a 24 V BLDC motor with Hall sensors.

The manufacturer specifies approximately 220 W continuous power, 0.729 N·m continuous torque, 2490 rpm nominal speed and 3170 rpm no-load speed.

Parameter	Specification
Motor	Maxon EC 90 Flat
Nominal voltage	24 V
Continuous power	220 W
Continuous torque	0.729 N·m
Nominal speed	2490 rpm
No-load speed	3170 rpm
Continuous current	9.43 A
Feedback	Hall sensors

Motor diameter	90 mm
Motor mass	Approximately 624 g

The Maxon GP 52 C is available with a 19 : 1 reduction and a maximum continuous torque of 15 N·m. Its maximum efficiency is approximately 83%.

Using this combination:

$$T_{out} = 0.729(19)(0.83) \quad (57)$$

$$\boxed{T_{out} \approx 11.5 \text{ N m}} \quad (58)$$

This is very close to the preliminary design target of 11.9 N·m.

At the maximum wheel speed:

$$N_m = 159.2(19) \quad (59)$$

$$\boxed{N_m \approx 3025 \text{ rpm}} \quad (60)$$

which is below the EC 90's no-load speed of approximately 3170 rpm.

The calculated continuous output torque is slightly below the conservative design target. Consequently, this combination should be considered a near-compliant candidate requiring detailed operating-point verification, rather than a fully validated final selection.

Advantages:

- Higher continuous torque than the EC 60;
- 220 W continuous power;
- 19 : 1 gearbox provides an appropriate motor speed range;
- GP 52 C 19:1 provides up to 15 N·m continuous gearbox output torque;
- Industrial planetary gearbox architecture;
- Compatible with the EPOS4/CANopen motion-control architecture.

Disadvantages:

- Larger 90 mm diameter;
- Higher mass than the EC 60;
- More expensive and mechanically more demanding;
- The calculated continuous torque is slightly below the conservative 11.9 N·m design target.

3.3 Candidate 3: Maxon EC 90 Flat 360 W

As a higher-performance alternative, the EC 90 Flat family also includes a 360 W version with Hall sensors. This motor provides approximately 0.961 N·m continuous torque and approximately 3210 rpm no-load speed at 24 V.

When combined with the GP 52 C 19:1 gearbox:

$$T_{out} = 0.961(19)(0.83) \quad (61)$$

$$\boxed{T_{out} \approx 15.2 \text{ N m}} \quad (62)$$

This exceeds the preliminary design requirement of approximately 11.9 N·m per wheel. The motor speed at the maximum vehicle velocity is:

$$N_m = 159.2(19) \quad (63)$$

$$\boxed{N_m \approx 3025 \text{ rpm}} \quad (64)$$

which remains below the motor's approximately 3210 rpm no-load speed.

The combination therefore provides a greater torque margin than the 220 W EC 90 configuration.

Advantages:

- Approximately 15.2 N·m estimated continuous gearbox output torque;
- Significant torque margin above the preliminary requirement;
- 360 W motor power provides a larger thermal and dynamic margin;
- Suitable motor speed with the 19 : 1 gearbox;
- Compatible with the EPOS4 BLDC/CANopen architecture.

Disadvantages:

- Larger and heavier motor;
- Higher cost;
- 360 W is potentially excessive for the current preliminary requirements;
- Requires careful thermal, mechanical, and electrical integration.

4 Comparison of Drivetrain Candidates

Configuration	Motor Power	Motor Torque	Approx. Wheel Torque	Preliminary Result
EC 45 + GP 42C 33:1	70 W	0.128 N·m	$\approx 3.6 \text{ N·m}$	Not suitable

EC 60 200 W + GP 52C 26:1	200 W	0.540 N·m	≈ 11.7 N·m	Promising
EC 90 220 W + GP 52C 19:1	220 W	0.729 N·m	≈ 11.5 N·m	Near target
EC 90 360 W + GP 52C 19:1	360 W	0.961 N·m	≈ 15.2 N·m	Strong candidate

5 Compatibility with EPOS4 and CANopen

The proposed drivetrain architecture uses the EPOS4 as the dedicated motor controller between the STM32 embedded system and the BLDC motor.

The communication chain is:

$$\boxed{\text{ROS2} \rightarrow \text{Jetson} \rightarrow \text{STM32} \rightarrow \text{CANopen} \rightarrow \text{EPOS4} \rightarrow \text{BLDC Motor}} \quad (65)$$

The EPOS4 implements the CANopen application layer and CiA 402 drive profile and supports operating modes such as Profile Velocity Mode, Cyclic Synchronous Velocity, Cyclic Synchronous Position, and Cyclic Synchronous Torque.

For BLDC operation, Hall sensors are required for commutation, while an incremental encoder is recommended when accurate velocity or position control is required.

Consequently, changing the motor does not require abandoning the current CANopen development work. The motor-specific parameters, sensor configuration, and EPOS4 commissioning parameters will need to be updated, but the overall STM32–CANopen–EPOS4 software architecture can remain.

6 Preliminary Recommendation

Based on the current assumptions, the initially selected EC 45 Flat 70 W with the GP 42C 33:1 gearbox should not be retained as the final drivetrain selection. The combination does not provide sufficient continuous torque or power for the preliminary 130 kg design case.

Three more suitable configurations have been identified:

1. **EC 60 Flat 200 W + GP 52 C 26:1** – compact and relatively lightweight solution, with an estimated output torque close to the preliminary requirement. This is a promising candidate for a compact AMR design, but the exact operating point and continuous gearbox rating must be verified.
2. **EC 90 Flat 220 W + GP 52 C 19:1** – provides more continuous torque and power while maintaining a suitable motor speed. It is close to the preliminary torque target and offers a reasonable balance between performance and size.
3. **EC 90 Flat 360 W + GP 52 C 19:1** – provides the largest performance margin and approximately 15.2 N·m estimated continuous output torque. However, it is larger, heavier, and potentially over-dimensioned for the current preliminary requirements.

The **EC 60 200 W** configuration should therefore be investigated as the compact option, while the **EC 90 220 W** can be retained as a balanced alternative and the **EC 90 360 W** as a higher-performance alternative.

The final motor and gearbox selection should be made only after checking the complete motor speed–torque curve, gearbox continuous/intermittent torque ratings, thermal conditions, wheel mounting arrangement, and the exact EPOS4 controller variant.

7 Motor Controller and CANopen Interface

7.1 Selected Motor Controller: Maxon EPOS4 Module 50/15 CAN

The Maxon EPOS4 Module 50/15 CAN is selected as the motor controller between the STM32 embedded controller and the BLDC drive motors. The EPOS4 is a digital positioning controller designed for brushed DC motors with encoders as well as brushless EC motors (BLDC) equipped with Hall sensors and encoders. It supports motor applications up to 750 W of continuous output power, making it suitable for the motor candidates considered in this study.

The controller operates from a 10–50 V DC supply and provides a maximum continuous output current of 15 A. It communicates through a CAN interface and implements the CANopen CiA 402 drive profile. The EPOS4 is therefore used as the CANopen slave device, while the STM32 acts as the higher-level CANopen master responsible for commanding and monitoring the motor controller.

Parameter	Specification
Component	Maxon EPOS4 Module 50/15 CAN
Function	Digital motor controller responsible for BLDC commutation, current, velocity and position control, motor feedback processing, and execution of motion commands received through CANopen.
Motor Compatibility	Brushless EC motors (BLDC) with Hall sensors and encoders; brushed DC motors with encoders.
Supply Voltage	10–50 V DC
Maximum Continuous Output Power	750 W
Maximum Continuous Output Current	15 A
Communication Interface	CAN / CANopen
CANopen Profile	CiA 402 – Device Profile for Drives and Motion Control
CAN Interface Standard	ISO 11898-2
Maximum CAN Bit Rate	1 Mbit/s
CANopen Node Type	Slave / Motion Control Slave
Feedback	Hall sensors and incremental encoder
Quantity Required	2 – one controller for each drive motor

The EPOS4 is not intended to operate as an independent high-level motion controller. It operates as a CiA 402 motion-control slave and therefore requires a higher-level master to issue commands. In the proposed architecture, this role is assigned to the STM32 embedded controller, which communicates with the EPOS4 through CANopen.

7.2 Common Controller for the Selected Motor Candidates

The EPOS4 controller is common to all three drivetrain candidates retained in the preliminary selection:

- Maxon EC 60 Flat 200 W + GP 52 C 26:1;
- Maxon EC 90 Flat 220 W + GP 52 C 19:1;
- Maxon EC 90 Flat 360 W + GP 52 C 19:1.

All three candidates are 24 V brushless motor configurations with Hall sensor feedback. The EPOS4 therefore provides a common motor-control and CANopen interface, allowing the embedded software architecture to remain largely unchanged if the final motor selection is modified.

The resulting control architecture is:

$$\boxed{\text{ROS 2} \rightarrow \text{Jetson} \rightarrow \text{STM32} \rightarrow \text{CANopen} \rightarrow \text{EPOS4} \rightarrow \text{BLDC Motor}} \quad (66)$$

The motor-specific parameters, such as nominal current, torque constant, speed, Hall sensor configuration, encoder resolution, gearbox ratio and velocity limits, must be configured according to the final selected motor. However, the communication layer between the STM32 and the EPOS4 remains based on the same CANopen/CiA 402 architecture.

7.3 EPOS4 and CANopen Communication

The EPOS4 communicates with the STM32 using the CAN bus physical layer and the CANopen protocol. The EPOS4 is configured as a CANopen slave node, while the STM32 acts as the CANopen master.

The STM32 is responsible for sending motion commands and receiving status information from the EPOS4. Typical information exchanged includes:

- Motor enable and disable commands;
- Target velocity;
- Target position, when applicable;
- Actual motor velocity;
- Actual position;
- Status word;
- Control word;

- Motor current and torque-related information;
- Fault and warning information;
- Emergency and error messages.

The CAN interface of the EPOS4 supports communication at up to 1 Mbit/s. The CAN bus should be terminated with 120 Ω resistors at both physical ends of the bus.

7.4 Electronic Data Sheet (EDS)

An Electronic Data Sheet (EDS) is used to describe a CANopen device and its supported object dictionary. It contains information such as object types, data types, access permissions, PDO mapping, default values and permitted value ranges.

For the EPOS4, the EDS is particularly useful during the development of the STM32 CANopen master because it provides the information required to identify and configure the CANopen objects exposed by the motor controller.

However, the EPOS4 EDS cannot be selected solely from the motor model. According to Maxon, the content of the EDS depends on the exact EPOS controller hardware type and firmware version. Maxon therefore does not provide one generic EPOS4 EDS file as a website download.

The EDS must instead be exported from the actual controller using *EPOS Studio*. The procedure is:

1. Connect the EPOS4 controller to EPOS Studio.
2. Open the *Object Dictionary* tool.
3. Right-click within the Object Dictionary.
4. Select *Export EDS File*.
5. Save the generated `.eds` file for use during CANopen development.

The exported EDS corresponds to the hardware and firmware version of the connected EPOS4 and should therefore be used as the reference file for the STM32 CANopen implementation.

EDS Status: *To be exported from the selected EPOS4 Module 50/15 CAN once the physical controller and its firmware version are available.*

7.5 Impact on Embedded Software Development

The use of the EPOS4 as the CANopen motor controller allows the STM32 firmware to be divided into two main layers.

- **CANopen communication layer:** responsible for CAN frames, Network Management (NMT), Service Data Objects (SDO), Process Data Objects (PDO), heartbeat monitoring and emergency messages.
- **Motor-control interface:** responsible for translating high-level velocity or position commands received from the robot control software into the appropriate CiA 402 commands and monitoring the EPOS4 status and feedback.

The intended embedded architecture is therefore:

$$\boxed{\text{STM32 Application} \rightarrow \text{Motor Interface} \rightarrow \text{CANopen Stack} \rightarrow \text{CAN Peripheral} \rightarrow \text{EPOS4}} \quad (67)$$

This architecture makes the STM32 independent from the exact motor model at the application level. When the final motor is selected, only the motor-specific configuration and scaling parameters need to be updated, while the fundamental CANopen communication architecture can be retained.